

Calculus - Integral Calculus

Feladat 1. Calculate the indefinite integral of the following functions!

$$\begin{aligned} (1) \int x^2 \sin(x) dx, \quad (2) \int x \sin(x^2) dx, \quad (3) \int \frac{2}{x+2} dx, \quad (4) \int x \cdot \sqrt[4]{x^2+3} dx, \\ (5) \int \cos(2x) \sin(x) dx, \quad (6) \int \sqrt[4]{x+3} dx, \quad (7) \int \frac{1}{\cos(x) \sin(x)} dx, \quad (8) \int \ln(x) dx, \\ (9) \int \sqrt{x^2+3} dx, \quad (10) \int 5^{2x+1} dx, \quad (11) \int x^2 \ln(x^3+1) dx, \quad (12) \int x^2 \ln(x^3) dx, \\ (13) \int (x+1)^3 \cosh(x) dx, \quad (14) \int \frac{x+3}{(x^2+6x+9)^5} dx, \quad (15) \int \frac{1}{x^3} dx, \\ (16) \int \sqrt{1-x^2} dx, \quad (17) \int \sqrt{2x^2+7} dx \quad (18) \int \cos^5(x) \sin(x) dx, \quad (19) \int \frac{9^x}{1-3^x} dx, \\ (20) \int x \ln(x^2+3) dx \quad (21) \int \frac{\sin(2x)}{\sin^2(x)} dx, \quad (22) \int (x^2+4) \cos^2(x) dx \end{aligned}$$

Feladat 2. Calculate the following Riemann integrals!

$$\begin{aligned} (1) \int_0^1 x^2 dx, \quad (2) \int_0^{2\pi} \sin(x) dx, \quad (3) \int_{-1}^1 x^3 + 1 dx, \quad (4) \int_0^{\pi/2} (x^2+2) \cos(2x) dx, \\ (5) \int_1^2 \frac{2}{3x+1} dx, \quad (6) \int_0^1 x \ln(x^2+2x+1) dx, \quad (7) \int_0^1 2^{x+1} \sin(x) dx \end{aligned}$$

Feladat 3. Calculate the following improper integrals!

$$\begin{aligned} (1) \int_0^1 \frac{1}{x} dx, \quad (2) \int_0^{+\infty} \frac{1}{x^3} dx, \quad (3) \int_{-\pi/2}^{\pi/2} \tan(x) dx, \quad (4) \int_0^{+\infty} e^{1-x} dx, \\ (1) \int_0^1 \frac{1}{x} dx, \quad (2) \int_0^{+\infty} \frac{1}{x^3} dx, \quad (3) \int_{-\pi/2}^{\pi/2} \tan(x) dx, \quad (4) \int_0^{+\infty} e^{1-x} dx, \end{aligned}$$